DnD Web Based Character Sheet

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Team 10 :

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Introduction

- The project acts as a web based Dungeons & Dragons character sheet while storing and retrieving character information from a database
- Our target audience is people already interested in Dungeons & Dragons between the ages of 12 - 18
- The importance of our project is to introduce this growing demographic (Fiction Horizon, 2024) to the necessary basics of computer science and more (European Education Area)
- The project objectives are :
  - Product is suitable replacement for physical character sheets
  - Include features that introduce computer science basics
- Our objectives requirements are :
  - Functioning product that is able to store and retrieve data
  - Account system allowing user only access to their characters

Software Engineering

The software development strategy approach our process most aligned with was Scrum. This is seen as we assigned each of the team members roles such as scrum master, product owner, software lead & Etc as well as organising reoccurring meetings. We chose this strategy because research (Atlassian, 2026) demonstrated it was both a widely used and affective strategy across the industry. One software engineering decision we chose to made was adapting for our team "scrums" from the industry standard daily to a more achievable weekly meeting over a group voice chat. This was done to make the module less overwhelming as we still had other modules and having calls online made it easier for each team member to attend the meetings.

This decision proved to be a success as it confirmed that we had made the right choice of software development strategy with our team members staying productive while also specifying their working to what was necessary for their roles assigned by the scrum strategy.

We used version control to ensure our code was separated by each team members role and stage of development. This can be seen in our repository (GitHub, 2026) on GitHub for this project where each issue was assigned to the relevant member, and branched from main. This let us ensure we had efficient risk management and evidence of each of our team member's contributions to the code. Reflecting on our strategy, we realised alternate approaches we could have taken with one being the software development strategy spiral. This is due to spiral focusing on preventing errors in code as our project grew larger, the risk of errors would become greater.

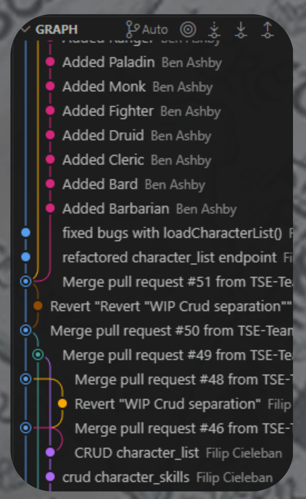

Figure 1: Screenshots demonstrating how our software was constructed using version control

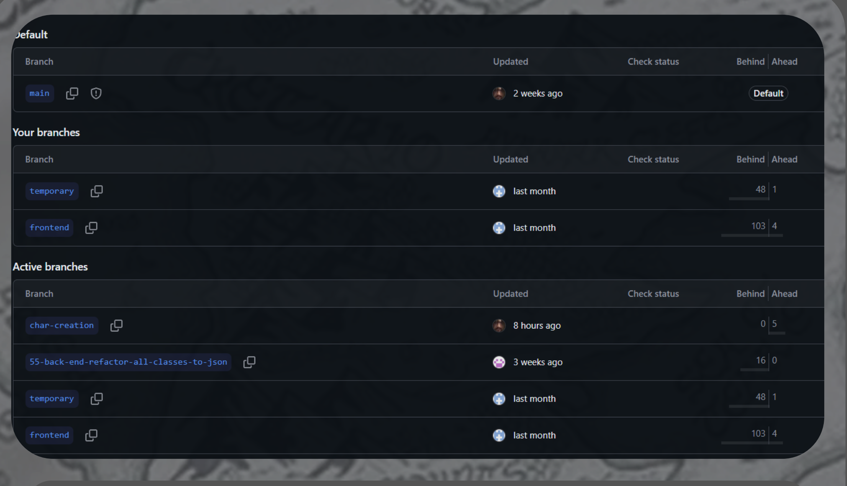

Figure 2: "main" code was a dependency for our separated branches

Implementation

The artefact was implemented as a full-stack web application that integrates a front-end command-line user interface, back-end API , and relational database to deliver a D&D character sheet system.

The front-end was developed using HTML,CSS and JavaScript, providing users with an interface to interact with. The design prioritised simplicity compared to the industry standard to ensure accessibility for the unskilled target audience.

The back-end was built using FastAPI for handling requests from the front-end, processing data and managing communication with the database through defined API endpoints. The system implements an account-based structure, ensuring that users can only access and modify their own characters for security purposes.

A MySQL database using the InnoDB storage engine was used to store user profiles and character data. This ensures reliable data retrieval, allowing users to return and continue editing their characters. The application was then containerised using docker as this allowed the whole system to be run using a single command.

Evaluation

Looking back at our groups main goals for this artefact, we were able to achieve our first objective of creating a product more than suitable replacing the standard paper character sheets. This is proven by the fact we were able to create a functioning artefact with an account system meaning only users who created a character could access them. This along with the ability to create as many characters as you want vastly outweighs any possible benefit of using paper character sheets.

The other goal we set ourselves was to have this artefact introduce basic computer science concepts to the user and target audience. We were able to achieve this a multitude of times with one such time being our user interface as its a command-line interface similar to a computer terminal. Another example of this is the account-based structure as it's a common feature among programs yet explained in detail to the user here.

Overall, we were incredibly successful in achieving a high standard in our functionality, usability and teamwork while also meeting both our goals and requirements for the project too. However had we had more time, we could have potentially improved the look of the command-line interface to match their typical design and layout in the industry.

Legal, Ethical, Social & Professional Issues

While coding our artefact, we respected intellectual property rights (Gov.UK, 2023) and followed the "UK Copyright and Right in Databases Regulations 1997" by choosing to develop our artefact entirely from scratch. Not using third-party libraries for resources, we avoided the legal risks using copyrighted material or violating licenses. we also ensured that all development environments and software tools used to build our project were operated under appropriate educational or open-source agreements. We acknowledged the "UK Equality Act 2010" (Legislation Gov.UK, 2010) by ensuring our software doesn't treat people with protected characteristics unfairly. By collecting no information other than username and email, there is little identifiable information on this front. Additionally, it lessens our burden as a data controller for GDPR.

The development of our D&D character creator was influenced by the framework of Virtue Ethics as this framework is about the motivation and virtues of our team as developers. Our aim was to encourage creativity and build a useful tool to lower the barriers to entry for the tabletop RPG community, making a mathematically complicated game more approachable and less complicated. We wanted to build a program that is supportive and provides a positive experience for the user, not designing a system for engagement or harvesting the user's data. Our virtues were honesty, inclusivity and community support, so we actively avoided deceptive design patterns. For example, our user interface is completely transparent and free from psychological manipulation like 'confirmshaming', misleading information, or hidden subscriptions and we have a strong duty of care to our users.

A key social issue addressed by this project is the decline in digital literacy (European Education Area, 2024) by children ages 12 - 18. Modern technology often prioritises ease of use through touch-based and highly automated system interfaces, which reduces exposure to how systems actually function. To address this problem, the project introduces programming concepts through a familiar and engaging medium being D&D. This approach then helps make computer science more accessible, helping tackle this social issue.

The professional issue we encountered when developing our project was a lack of communication and work from team members. At the time we acknowledged this by redistributing work load and contacting our project supervisor. Remaining respectful and patient, we eventually heard back from the participants and carried on with the project as a team.

Group Work Reflection

Having individual self reflections, both Ben and Lorenzo though they cooperated effectively on the back-end yet they said they should've tested their code more between commits. Filip and Alex both said their leadership on the artefact and poster respectively made sure work was done well yet taking on too much work caused Filip to burn out while Alex admitted to needing to check on others work more often. Arun said his work on the poster was high-quality and Lucas thought his speed working was effective yet they both admitted to needing to attend more scrum team meetings.

Overall, most of our team was involved and cohesive in the development of the D&D character spreadsheet. We set up clear communication protocols using Discord for weekly scrum meetings and a shared Gantt chart to map out the work breakdown structure and critical dependencies in sprint 0. We would try to keep our sprint reviews honest and as reflective as possible, which would allow us to pivot our resources where they were needed while still delivering the core functionality of the application on time.

However, our Gantt chart shows the core sprint 2 development tasks (front-end, back-end and database) running perfectly parallel from Jan 26th to Feb 9th. Actually, these were "end-to-start" dependencies and held up the next stage of "Data Flow Creation" to follow immediately because of delays in finalising the database schema. We had also given ourselves a generous three week window (March 30 – April 19) for "Bug fixing and optimisation" on our Gantt chart. However, this task being completely isolated in the final "Finalisation" sprint created an unnecessary bottleneck. We did not address technical debt as we went along with the development phase, which then meant we started sprint 4 with a large backlog of errors to fix before our April 20 "Stable Release Version" milestone.

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